OEE Tracker

Step-by-step guide: track Overall Equipment Effectiveness by machine and shift

What it does

You log one row per machine per shift. The workbook calculates Availability, Performance, Quality and OEE for every row, then summarizes each machine on the **Dashboard** for any date range you choose, with a chart and a comparison to your target.

Yellow cells: type here.	Grey cells: calculated. Don't type over them.

The sheets

Sheet	What it's for
How to use	A one-page summary.
Setup	Your machines and each one's ideal cycle time. Up to 20 machines.
Production Log	One row per machine per shift. Up to 500 rows.
Dashboard	Totals and OEE by machine for a date range, plus a chart.

Before you start

1. Open the file in your spreadsheet app (see **Works with** in the listing). If Excel shows a yellow *Protected View* bar because the file was downloaded, click **Enable Editing**.
2. Save a copy under your own name (for example, add your plant or year) and keep the original download as a clean spare.
3. Read the **How to use** sheet. It's a one-page summary of this guide.

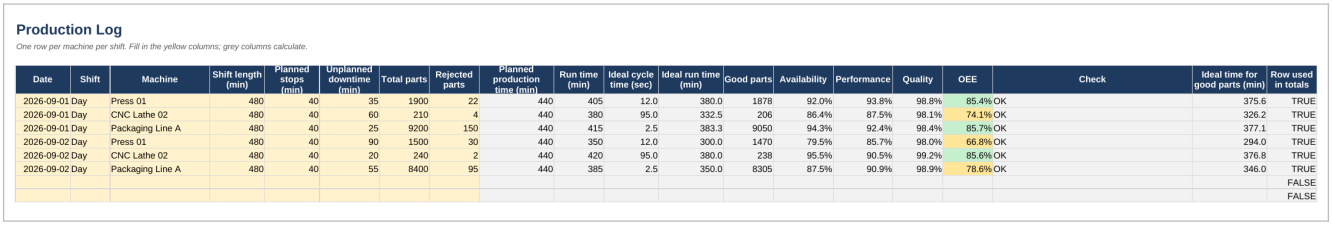
Step 1: Set up your machines

1. Go to the **Setup** sheet.
2. In **Machine**, type each machine's name once, one per row (up to 20). Use the names your team already uses.
3. In **Ideal cycle time (seconds per part)**, enter the fastest time the machine can sustainably make one good part. For a line, use the time per finished unit at rated speed. Example: a machine rated at 300 parts per hour has an ideal cycle time of 3600 / 300 = **12 seconds**.
4. Type each machine only once. If a name is repeated, **Check** on the repeat says **Duplicate: only the first row is used**; delete the repeat.
5. Clear the three EXAMPLE machines when you no longer need them (yellow cells only).

Getting the ideal cycle time right matters most: if it's too slow, Performance can show over 100% and the Check column will flag it.

Step 2: Log each shift

1. Go to the **Production Log** sheet.
2. Fill in one row per machine per shift. The yellow columns are: **Date**, **Shift** (any label, e.g. Day or 1st), **Machine** (pick from the drop-down), **Shift length (min)**, **Planned stops (min)**, **Unplanned downtime (min)**, **Total parts** and **Rejected parts**. Shift length and total parts are required; blank planned stops, downtime or rejects count as 0.
3. **Planned stops** are times the machine isn't expected to run, like breaks and scheduled meetings. **Unplanned downtime** is every other minute it was stopped: breakdowns, changeovers, waiting for material.
4. The grey columns fill in by themselves, including **OEE** (coloured green, amber or red) and **Check**.
5. Look at **Check** on each new row. It should say **OK**. If not, see the table below. Rows with an error are left out of the Dashboard until you fix them. The "performance above 100%" warning is not an error: those rows still count.
6. The rows marked **EXAMPLE** show the expected format. When you're ready, select just their **yellow** cells and press **Delete** to clear them. Don't delete whole rows: that removes the formulas in the grey columns too.



Production Log

One row per machine per shift. Fill in the yellow columns; grey columns calculate.

Date	Shift	Machine	Shift length (min)	Planned stops (min)	Unplanned downtime (min)	Total parts	Rejected parts	Planned production time (min)	Run time (min)	Ideal cycle time (sec)	Ideal run time (min)	Good parts	Availability	Performance	Quality	OEE	Check	Ideal time for good parts (min)	Row used in totals
2026-09-01	Day	Press 01	480	40	35	1900	22	440	405	12.0	380.0	1878	92.0%	93.8%	98.8%	85.4%	OK	375.6	TRUE
2026-09-01	Day	CNC Lathe 02	480	40	60	210	4	440	380	95.0	332.5	206	86.4%	87.5%	98.1%	74.1%	OK	326.2	TRUE
2026-09-01	Day	Packaging Line A	480	40	25	9200	150	440	415	2.5	383.3	9050	94.3%	92.4%	98.4%	85.7%	OK	377.1	TRUE
2026-09-02	Day	Press 01	480	40	90	1500	30	440	350	12.0	300.0	1470	79.5%	85.7%	98.0%	66.8%	OK	294.0	TRUE
2026-09-02	Day	CNC Lathe 02	480	40	20	240	2	440	420	95.0	380.0	238	95.5%	90.5%	99.2%	85.6%	OK	376.8	TRUE
2026-09-02	Day	Packaging Line A	480	40	55	8400	95	440	385	2.5	350.0	8305	87.5%	90.9%	98.9%	78.6%	OK	346.0	TRUE
																			FALSE
																			FALSE

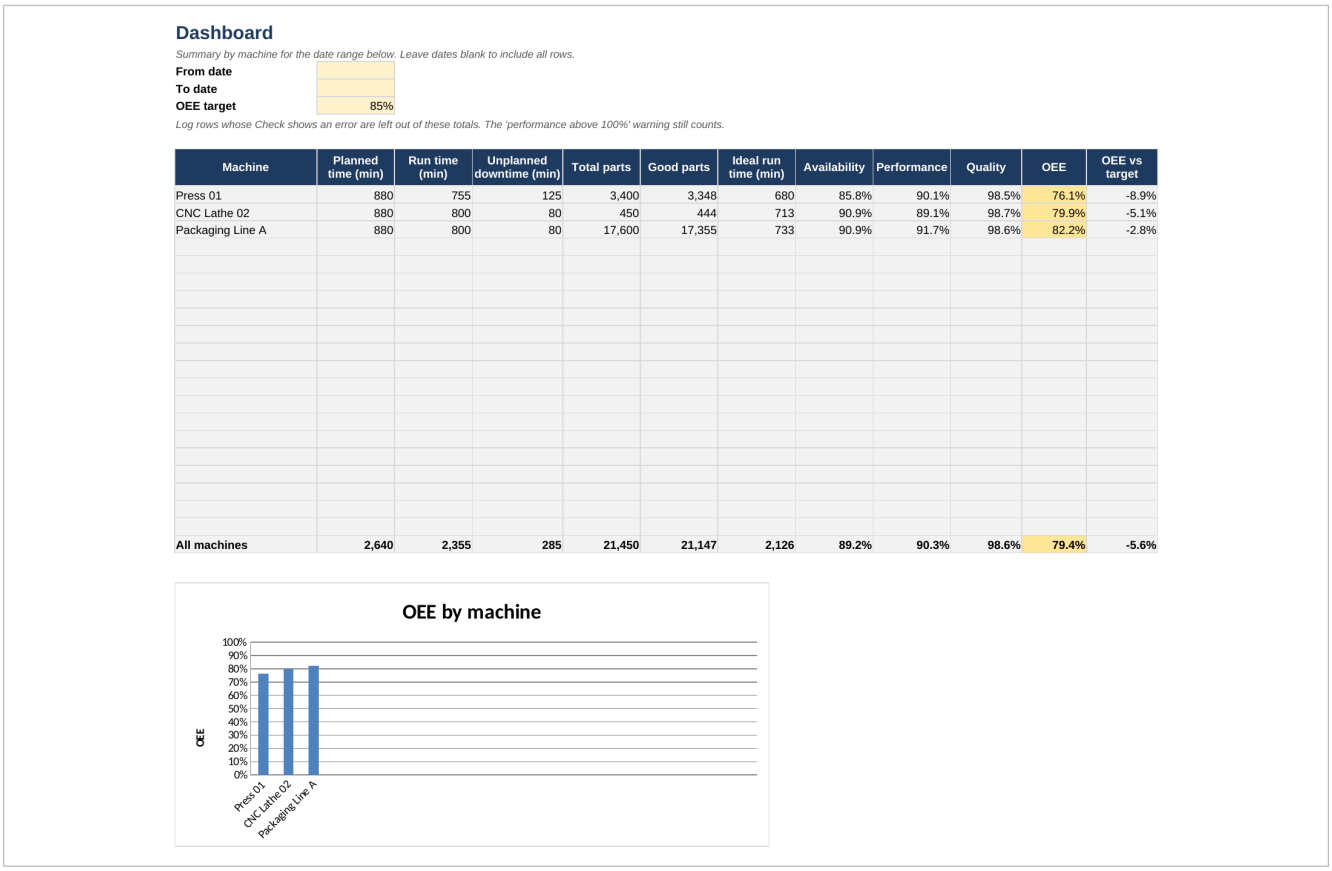
Production Log with the example rows. Yellow columns are typed in; grey columns calculate.

What the Check column means

Message	What to do
Machine not in Setup	The machine name isn't on the Setup sheet, or is spelled differently. Pick it from the drop-down, or add it to Setup.
Set an ideal cycle time in Setup	The machine has no ideal cycle time (or 0) on the Setup sheet.
Enter a date	The Date cell is empty or isn't a real date. Undated rows can't be placed in a date range, so they are left out.
Enter numbers: shift length and total parts are required	A required cell is empty, or a cell contains text instead of a number.
Values must be 0 or more (shift length above 0)	A negative number, or a shift length of 0.
Rejects > total parts	Rejected parts must be part of the total. Fix one of the two numbers.
Planned stops >= shift length	Planned stops leave no time to run. Check both numbers.
Downtime > planned time	Unplanned downtime is longer than shift length minus planned stops. Check the three time columns.
Check: performance above 100%	A warning only; the row still counts. The machine made parts faster than its ideal cycle time allows. Usually the cycle time in Setup is too slow, or the count is wrong.
OK	Nothing obviously wrong.

Step 3: Read the Dashboard

1. Go to the **Dashboard** sheet.
2. Optional: enter a **From date** and/or **To date**. Leave them blank to include every row. Both dates are included in the range.
3. Set your **OEE target** (85% is filled in). OEE cells turn **green** at or above target, **amber** from 60% up to the target, and **red** below 60%. Red always wins: below 60% is red even if your target is lower than 60%. **OEE vs target** shows the gap.
4. Each row adds up times and part counts first, then calculates the percentages, so a long shift counts more than a short one. **All machines** OEE is total ideal time for good parts divided by total planned production time, which stays correct when machines have different cycle times. Its **Quality** is weighted by ideal time (ideal time for good parts ÷ ideal time for all parts), so Availability × Performance × Quality still equals OEE.



Dashboard

Summary by machine for the date range below. Leave dates blank to include all rows.

From date

To date

OEE target 85%

Log rows whose Check shows an error are left out of these totals. The 'performance above 100%' warning still counts.

Machine	Planned time (min)	Run time (min)	Unplanned downtime (min)	Total parts	Good parts	Ideal run time (min)	Availability	Performance	Quality	OEE	OEE vs target
Press 01	880	755	125	3,400	3,348	680	85.8%	90.1%	98.5%	76.1%	-8.9%
CNC Lathe 02	880	800	80	450	444	713	90.9%	89.1%	98.7%	79.9%	-5.1%
Packaging Line A	880	800	80	17,600	17,355	733	90.9%	91.7%	98.6%	82.2%	-2.8%
All machines	**2,640**	**2,355**	**285**	**21,450**	**21,147**	**2,126**	**89.2%**	**90.3%**	**98.6%**	**79.4%**	**-5.6%**

Dashboard: totals and OEE by machine, the all-machines total, and the OEE chart.

How the numbers are calculated

Measure	Formula
Planned production time	Shift length - planned stops
Run time	Planned production time - unplanned downtime
Availability	Run time ÷ planned production time
Performance	(Ideal cycle time × total parts) ÷ run time
Quality	Good parts ÷ total parts (good parts = total - rejected)
OEE	Availability × Performance × Quality = (ideal cycle time × good parts) ÷ planned production time

85% OEE is a commonly cited "world-class" benchmark. Many plants start well below it; improving against your own baseline matters more.

Questions

The log is full (500 rows). What now? Save the workbook as a new file for the next period (for example, one per quarter), clear the yellow cells in the log, and keep going.

Can I add more than 20 machines? Not in this version. Message us if you need more.

Can I rename the sheets or move columns? Please don't. The formulas refer to them by position.

Need help?

Message us through Etsy (the **Message seller** button on our shop or your order). Tell us which file, which sheet, and what you typed. We answer questions about using the templates as described in this guide; we can't do custom changes or set-up work.

These templates were designed with help from AI tools and checked by the seller.

Downtime Log & Pareto

Step-by-step guide: log every stop and see which causes cost the most time

What it does

You record each stop with a reason. The **Pareto** sheet ranks the reasons by minutes lost, with each reason's share and the running total, so you can see the few causes behind most of your downtime. Filter by date range and machine.

Yellow cells: type here.	Grey cells: calculated. Don't type over them.

The sheets

Sheet	What it's for
How to use	A one-page summary.
Setup	Your machines (up to 30) and your downtime reasons with categories (up to 30).
Downtime Log	One row per stop. Up to 1,000 rows.
Pareto	Reasons ranked by minutes lost, with a Pareto chart of the top 12.
By Machine	Downtime minutes and stop counts per machine.

Before you start

1. Open the file in your spreadsheet app (see **Works with** in the listing). If Excel shows a yellow *Protected View* bar because the file was downloaded, click **Enable Editing**.
2. Save a copy under your own name (for example, add your plant or year) and keep the original download as a clean spare.
3. Read the **How to use** sheet. It's a one-page summary of this guide.

Step 1: Set up machines and reasons

1. Go to the **Setup** sheet.
2. Under **Machine**, list your machines, one per row.
3. Under **Downtime reason** and **Category**, edit the 12 starter reasons to match how your plant talks about stops. Keep the list short, about 10 to 15 reasons. Categories (Breakdown, Material, and so on) can be renamed too.
4. List each machine and each reason only once. A repeated name shows **Duplicate: only the first row is used** in the **Check** column next to it; delete the repeat.

Setup

Machines and downtime reasons used by the drop-downs.

Machine	Check
Press 01	
CNC Lathe 02	
Packaging Line A	

Downtime reason	Category	Check
Mechanical breakdown	Breakdown	
Electrical / controls fault	Breakdown	
Tooling failure	Breakdown	
Changeover / setup	Setup & adjustment	
Adjustment / tuning	Setup & adjustment	
Waiting for material	Material	
Waiting for upstream process	Material	
Quality hold / inspection	Quality	
No operator available	Labor	
Planned maintenance	Planned	
Meeting / training	Planned	
Other (describe in notes)	Other	

Setup: machines on the left, reasons and categories on the right.

Step 2: Log every stop

1. Go to the **Downtime Log** sheet and use one row per stop.
2. Enter **Date**, **Shift** and **Machine** (from the drop-down).
3. Enter **Start** and **End** as times, like **7:15** and **7:50**. A stop that crosses midnight, like 23:40 to 0:25, is handled automatically (45 minutes).
4. If you only know how long it lasted, leave Start and End blank and type the minutes in **Minutes (if no start/end)**. Typed minutes always win over start and end times.
5. Pick a **Reason** from the drop-down. **Duration**, **Category** and **Check** fill in by themselves. Check should say **OK**; otherwise it tells you what's missing, such as a date, a reason, or minutes that aren't a number of 0 or more.
6. Add **Notes** if useful, especially for "Other".
7. The rows marked **EXAMPLE** show the expected format. When you're ready, select just their **yellow** cells and press **Delete** to clear them. Don't delete whole rows: that removes the formulas in the grey columns too.

Downtime Log

One row per stop. Use start/end times, or type minutes if you only know the duration.

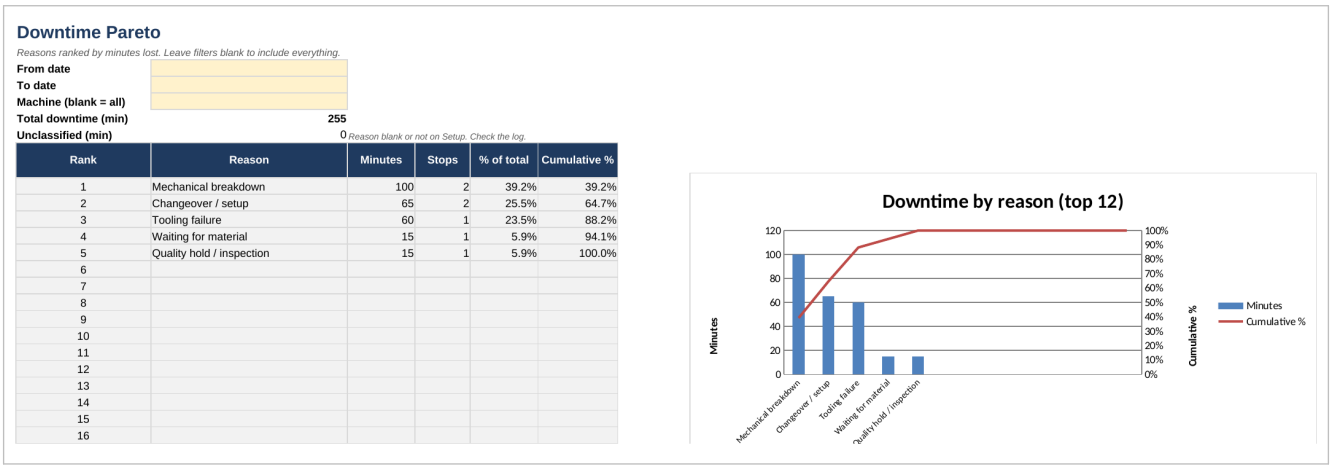
Date	Shift	Machine	Start (hh:mm)	End (hh:mm)	Minutes (if no start/end)	Duration (min)	Reason	Category	Notes	Check
2026-09-01	Day	Press 01	07:15	07:50		35	Changeover / setup	Setup & adjustment	EXAMPLE - Die change	OK
2026-09-01	Day	Press 01	10:05	10:20		15	Waiting for material	Material	EXAMPLE	OK
2026-09-01	Day	CNC Lathe 02	09:00	10:00		60	Tooling failure	Breakdown	EXAMPLE - Insert broke	OK
2026-09-01	Night	Packaging Line A	23:40	00:25		45	Mechanical breakdown	Breakdown	EXAMPLE - Conveyor jam, crossed midnight	OK
2026-09-02	Day	Press 01			55	55	Mechanical breakdown	Breakdown	EXAMPLE - Minutes typed directly	OK
2026-09-02	Day	Packaging Line A	13:00	13:30		30	Changeover / setup	Setup & adjustment	EXAMPLE	OK
2026-09-02	Day	CNC Lathe 02	14:10	14:25		15	Quality hold / inspection	Quality	EXAMPLE	OK

Downtime Log with the example rows, including a stop that crosses midnight.

If **Category** shows **Not in Setup**, the reason was typed instead of picked, or is spelled differently from the Setup list. Its minutes still count in the total, under **Unclassified** on the Pareto sheet. Pick the reason from the drop-down so it's counted under the right reason. A row counts as a **stop** when its duration is more than 0 minutes; rows without a date, or with invalid minutes, are left out of the totals.

Step 3: Read the Pareto

1. Go to the **Pareto** sheet.
2. Optional: set **From date**, **To date** and **Machine**. Leave any of them blank to include everything. Both dates are included in the range.
3. **Total downtime** is every valid logged minute in the range. **Unclassified** shows minutes whose reason is blank or not on the Setup list. Fix those in the log so they're ranked.
4. Read the table from the top: **Rank**, **Reason**, **Minutes**, **Stops**, **% of total** and **Cumulative %**. Reasons with no minutes in the range are left out.
5. The chart shows the top 12 reasons as bars, with the cumulative % as a line on the right-hand axis.
6. Start improvement work with the reasons at the top. Often the first two or three account for most of the lost time.



Downtime Pareto

Reasons ranked by minutes lost. Leave filters blank to include everything.

From date	
To date	
Machine (blank = all)	
Total downtime (min)	255
Unclassified (min)	0 *Reason blank or not on Setup. Check the log.*

Rank	Reason	Minutes	Stops	% of total	Cumulative %
1	Mechanical breakdown	100	2	39.2%	39.2%
2	Changeover / setup	65	2	25.5%	64.7%
3	Tooling failure	60	1	23.5%	88.2%
4	Waiting for material	15	1	5.9%	94.1%
5	Quality hold / inspection	15	1	5.9%	100.0%
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					

Pareto: reasons ranked by minutes, with cumulative % and the Pareto chart.

Step 4: Compare machines

The **By Machine** sheet shows total downtime, number of stops, average minutes per stop and each machine's share of all downtime, plus **No machine / not in Setup** and **All logged downtime** rows so the totals always add up. It uses the date range set on the Pareto sheet; the machine filter doesn't apply here.

Downtime by Machine

Uses the date range set on the Pareto sheet (machine filter is ignored here). A stop is any row with more than 0 min

Machine	Downtime (min)	Stops	Avg min per stop	% of all downtime
Press 01	105	3	35.0	41.2%
CNC Lathe 02	75	2	37.5	29.4%
Packaging Line A	75	2	37.5	29.4%

By Machine: downtime minutes, stops and share per machine.

Tips

- If **Other** climbs into your top reasons, read its notes and add the missing reason to Setup.
- Durations are rounded to whole minutes.
- Log stops the same day, while people remember the cause.

Questions

The log is full (1,000 rows). What now? Save the workbook as a new file for the next period, clear the yellow cells in the log, and keep going.

Can I rename the sheets or move columns? Please don't. The formulas refer to them by position.

Need help?

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Preventive Maintenance Schedule

Step-by-step guide: plan recurring maintenance and see what's due

What it does

You list each recurring maintenance task and how often it's due. Each time a task is done, you add a line to the **Completion Log**. The plan then shows each task's last completion, next due date, days left and a colour-coded status.

Yellow cells: type here.	Grey cells: calculated. Don't type over them.

The sheets

Sheet	What it's for
How to use	A one-page summary.
PM Plan	One row per asset and task, with status. Up to 300 tasks.
Completion Log	One row each time a task is done. Up to 2,000 rows.

Before you start

1. Open the file in your spreadsheet app (see **Works with** in the listing). If Excel shows a yellow *Protected View* bar because the file was downloaded, click **Enable Editing**.
2. Save a copy under your own name (for example, add your plant or year) and keep the original download as a clean spare.
3. Read the **How to use** sheet. It's a one-page summary of this guide.

Step 1: Build your plan

1. Go to the **PM Plan** sheet.
2. Add one row per asset and task. Fill in **Asset**, **Task**, **Frequency (days)** and **Assigned to**. The frequency is a fixed number of days: 1 = daily, 7 = weekly, 14 = every two weeks, 30 = about monthly, 90 = about quarterly, 365 = about yearly.
3. If you know when a task was last done before you started using this workbook, enter that date in **Last done before log**. Otherwise leave it blank: the task shows as due today, every day, until you log it or enter a date.
4. Keep each asset + task combination on one row only. A repeated combination shows **Duplicate row**; a blank, zero or negative frequency shows **Check frequency**. Both are counted under **Needs fixing** at the top right.

5 The rows marked **EXAMPLE** show the expected format. When you're ready, select just their **yellow** cells and press **Delete** to clear them. Don't delete whole rows: that removes the formulas in the grey columns too.

PM Plan

One row per asset and task.

Due soon window (days) 7

Overdue	1
Due soon	2
OK	2
Needs fixi	0

Asset	Task	Frequency (days)	Assigned to	Last done before log	Last completed	Next due	Days until due	Status	Notes
Air compressor 1	Change oil filter	90	Maintenance	2026-07-03	2026-07-03	2026-10-01	-10	OVERDUE	EXAMPLE
Press 01	Lubricate slides and gibs	7	Operator	2026-10-08	2026-10-08	2026-10-15	4	Due soon	EXAMPLE
Forklift 2	Inspect forks, chains and hydraulics	30	Maintenance	2026-09-14	2026-09-14	2026-10-14	3	Due soon	EXAMPLE
CNC Lathe 02	Clean chip conveyor and coolant filter	14	Operator		2026-10-09	2026-10-23	12	OK	EXAMPLE
Dust collector	Replace filter cartridges	180	Maintenance	2026-09-21	2026-09-21	2027-03-20	160	OK	EXAMPLE

PM Plan with example tasks, statuses and the counts box (top right).

Step 2: Log completed work

1 Go to the **Completion Log** sheet.

2 Each time a task is done, add a row: **Date done**, **Asset** and **Task** (pick both from the drop-downs so they match the plan exactly), **Done by** and any **Notes**.

3 Check **Matches plan?**. It should say **OK**. **Enter the date done** means the date is missing. **Fill in asset and task** means one of them is blank. **No matching asset + task in plan** means the asset or task is spelled differently from the PM Plan, so the plan won't update. Fix the spelling.

Completion Log

Add a row each time a task is completed. Pick asset and task from the drop-downs.

Date done	Asset	Task	Done by	Notes	Matches plan?
2026-10-09	CNC Lathe 02	Clean chip conveyor and coolant filter	J. Smith	EXAMPLE	OK

Completion Log: each completion updates the plan.

Step 3: Read the plan

1 Back on **PM Plan**, the grey columns update automatically: **Last completed** (the most recent of the logged dates and the before-log date), **Next due**, **Days until due** and **Status**.

2 **OVERDUE** (red) means the due date has passed. **Due soon** (amber) means it's due within the window. **OK** (green) means it's not due yet.

3 Change the **Due soon window (days)** at the top of the sheet. It starts at 7.

4 The counts box at the top right shows how many tasks are Overdue, Due soon, OK, and Needs fixing.

5 To get this week's work list, sort or filter by Status: **Data > Filter** in Excel.

Dates and statuses use today's date from your computer, so they change from day to day. The example rows are set relative to today for the same reason.

Questions

What if a task is done early? Log the actual date. The next due date counts from the last completion.

The Completion Log is full. Save the workbook as a new file, enter each task's most recent date in **Last done before log**, clear the yellow cells in the log, and keep going.

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